

Bylaw 2023

Fall 2025

Experiment M1 Report
Transmission and Reflection on Glass Lens

ECE 331 – Electromagnetic Waves (Microwaves Laboratory)

Dr. Ahmed Sarhan / Dr. Hussein Kotb

G2

Names	IDs
Kareem Assem Abdelkari	2300367
Abdelrahman Sherif Medhat	2300100
Basel Magdy Farghly	2301145
Hoda sameh labib hemdan	2300662

Introduction Objectives

Electromagnetic (EM) waves, including light, consist of time-varying electric and magnetic fields that oscillate perpendicular to the direction of propagation. When such a wave encounters a boundary between two different dielectric media (such as air and glass), the wave energy is split: part of it is reflected back into the first medium, and the rest is transmitted (refracted) into the second medium.

The primary objective of this experiment is to experimentally analyze these transmission and reflection phenomena. Specifically, the experiment aims to:

1. **Analyze Wave Propagation:** Understand how the angle of incidence affects the power distribution between reflected and transmitted waves.
2. **Study Polarization Effects:** Differentiate between the behavior of S-polarized (Perpendicular/TE) and P-polarized (Parallel/TM) waves at a dielectric boundary.
3. **Determine Refractive Index:** Utilize the Brewster angle phenomenon—where the reflection of P-polarized light drops to zero—to experimentally calculate the refractive index of the glass lens.

Theoretical Background

The Physics of Reflection and Refraction

The interaction of a plane wave with a boundary is governed by the difference in the refractive indices (n1 and n2) of the two media.

- Snell's Law: Describes the relationship between the angle of incidence (theta_i) and the angle of refraction (theta_t):

$$n_1 \sin(\theta_i) = n_2 \sin(\theta_t)$$

This law dictates how the transmitted ray bends as it enters a denser medium.

Polarization States

The reflection coefficient is heavily dependent on the "Polarization" of the wave, which refers to the orientation of the Electric Field vector. We analyze two distinct cases:

- **S-Polarization (Perpendicular):** The electric field is perpendicular to the plane of incidence.
- **P-Polarization (Parallel):** The electric field is parallel to the plane of incidence.

Fresnel Equations

The fraction of reflected power (R) and transmitted power (T) are given by the Fresnel coefficients.

For Normal Incidence (theta_i = 0), polarization does not matter, and reflection is simply:

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2$$

For **Oblique Incidence**, the coefficients differ for S and P waves :

- **S-Pol Reflection:** $R_s = \left| \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t} \right|^2$
- **P-Pol Reflection:** $R_p = \left| \frac{n_2 \cos\theta_i - n_1 \cos\theta_t}{n_2 \cos\theta_i + n_1 \cos\theta_t} \right|^2$

Brewster's Angle A critical phenomenon observed in this experiment is **Brewster's Angle** (theta_B). At this specific angle of incidence, the reflection coefficient for P-polarized light (Rp) becomes zero. All energy is transmitted, and no P-polarized light is reflected. This allows us to calculate the refractive index of the second medium using the relation:

$$\tan(\theta_B) = \frac{n_2}{n_1}$$

Experimental Procedure

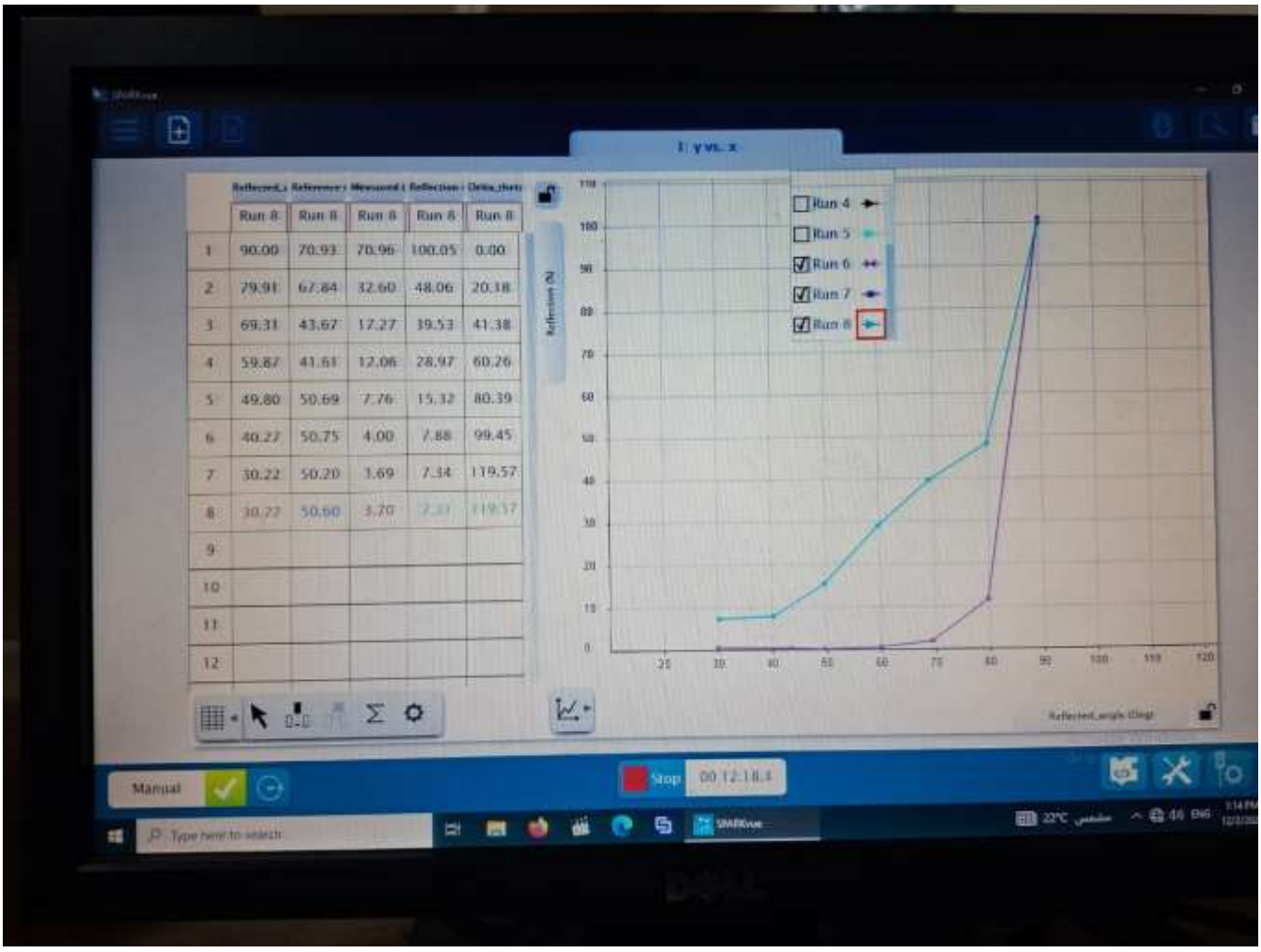
The experiment was conducted using a high-precision rotary table equipped with a D-lens (semi-cylindrical glass lens), a red diode laser source, and high-sensitivity light sensors.

1. **Setup Calibration:** The rotary table was aligned such that the laser beam struck the center of the D-lens. The system was calibrated to ensure the laser was centered on the photodetectors at normal incidence.
2. **Data Acquisition:** Using PASCO sensors and SparkVue software, the reflected light intensity was measured.
3. **Measurement Sweep:** The D-lens was rotated to vary the angle of incidence from 90degrees down to 30 degrees. At each angle, the reflected power was recorded.
4. **Polarization Control:** A linear polarizer was used to switch the incident light between S-polarization and P-polarization to observe the distinct reflection curves for each state.

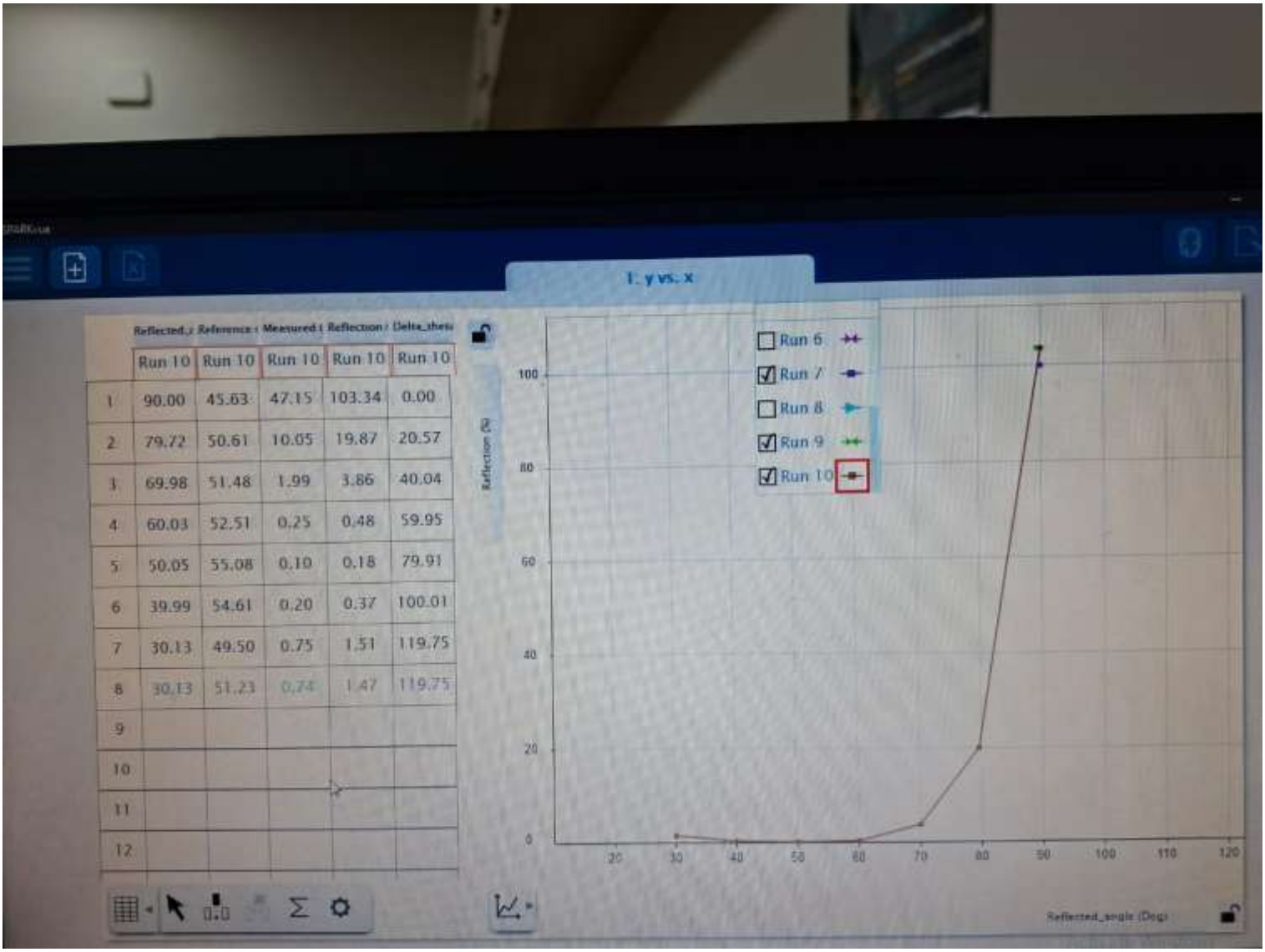
Experimental Results and Analysis

Measured Data The following table summarizes the reflected power measured at various angles of incidence for both S-polarized and P-polarized light.

S-polarized



P-polarized



	Run 10	Run 10	Run 10	Run 10	Run 10
1	90.00	45.63	47.15	103.34	0.00
2	79.72	50.61	10.05	19.87	20.57
3	69.98	51.48	1.99	3.86	40.04
4	60.03	52.51	0.25	0.48	59.95
5	50.05	55.08	0.10	0.18	79.91
6	39.99	54.61	0.20	0.37	100.01
7	30.13	49.50	0.75	1.51	119.75
8	30.13	51.23	0.74	1.47	119.75
9					
10					
11					
12					

Angle of Incidence (θi)	Reflected Power (S-Pol) [Run 8]	Reflected Power (P-Pol) [Run 10]
90.00°	100.05 %	103.34 %
79.9° / 79.7°	48.06 %	19.87 %
69.3° / 70.0°	39.53 %	3.86 %
59.9° / 60.0°	28.97 %	0.48 %
49.8° / 50.1°	15.32 %	0.18 % (Min)
40.3° / 40.0°	7.88 %	0.37 %
30.2° / 30.1°	7.34 %	1.51 %

Delta angle at angle of incidence 30° = 10.63

Q1) Is the measured delta angle the same as the angle of refraction? Why or why not? Use Snell's law to calculate the refractive index.

No, the measured "delta angle" delta = 10.63 is **not** the angle of refraction. It represents the **angle of deviation,** the difference between the incident angle and the refracted angle,

Verification: Using the measured refractive index $n_2 = 1.51$ at $\theta_i = 30^\circ$:

1. **Calculate θ_t:**

$$1.0 \times \sin(30^\circ) = 1.51 \times \sin(\theta_t)$$

$$\sin(\theta_t) = \frac{0.5}{1.51} \approx 0.3311$$

$$\theta_t = \sin^{-1}(0.3311) \approx 19.33^\circ$$

2. **Calculate Deviation (δ):**

$$\delta_{theoretical} = \theta_i - \theta_t = 30^\circ - 19.33^\circ = \mathbf{10.67}^\circ$$

Q2) Calculate the angle at which the reflected power becomes zero (Brewster's angle) and compare that with the measurements. Which polarization has a Brewster's angle?

Theoretical Calculation:
Using the measured refractive index n_2 = 1.51:

$$\theta_B = \tan^{-1}\left(\frac{n_2}{n_1}\right) = \tan^{-1}(1.51) \approx \mathbf{56.5}^\circ$$

Comparison with Measurements: Looking at the data for **Run 10 (P-Polarization)**:

- At $\theta = 50.05^\circ$, Reflection is 0.18%.
- At $\theta = 60.03^\circ$, Reflection is 0.48%. The minimum reflection occurs between 50° and 60°, which aligns with the theoretical value of 56.5°.

Which polarization? Only **P-Polarization (Parallel)** exhibits a Brewster angle. As seen in the graph for Run 10, the curve dips to near zero, whereas the S-Polarization (Run 8) curve never drops below 7%.

Q3) Plot the Reflected powers... Does it agree with expected results?

Yes, it agrees with the equations and graph.

S-Polarization (Run 8): Shows a steady increase in reflection as the angle of incidence increases, never reaching zero.

P-Polarization (Run 10): Clearly demonstrates the "dip" characteristic of the Brewster angle effect, where reflection nearly vanishes around 50-60 before rising sharply again at (>80).

Q4) MATLAB Program

Group Number is 2

MATLAB code

Editor - D:\reports\electromagnetics\M1\matlab_code.m

matlab_code.m

```
%              Exp 1: Q4 Solution
clc; clear; close all;

%    Define Range of Angles (0 to 90 degrees)
theta_deg = 0:0.1:90;
theta_i = deg2rad(theta_deg);

%    Define Refractive Indices
n1 = 1.0; % Air
n2 = 1.3; %           GROUP 2 VALUE (1 + 2*0.15)

%              Function for Fresnel Coefficients
% Calculates both Reflection (R) and Transmission (T)
calc_fresnel_RT = @(n1, n2, theta_i) deal( ...
    abs( (n1 .* cos(theta_i) - n2 .* sqrt(1 - (n1./n2 .* sin(theta_i)).^2)) ./ ...
         (n1 .* cos(theta_i) + n2 .* sqrt(1 - (n1./n2 .* sin(theta_i)).^2)) ).^2, ... % Rs
    abs( (n2 .* cos(theta_i) - n1 .* sqrt(1 - (n1./n2 .* sin(theta_i)).^2)) ./ ...
         (n2 .* cos(theta_i) + n1 .* sqrt(1 - (n1./n2 .* sin(theta_i)).^2)) ).^2  );  % Rp

%    Compute Reflection Coefficients (R)
[Rs, Rp] = calc_fresnel_RT(n1, n2, theta_i);

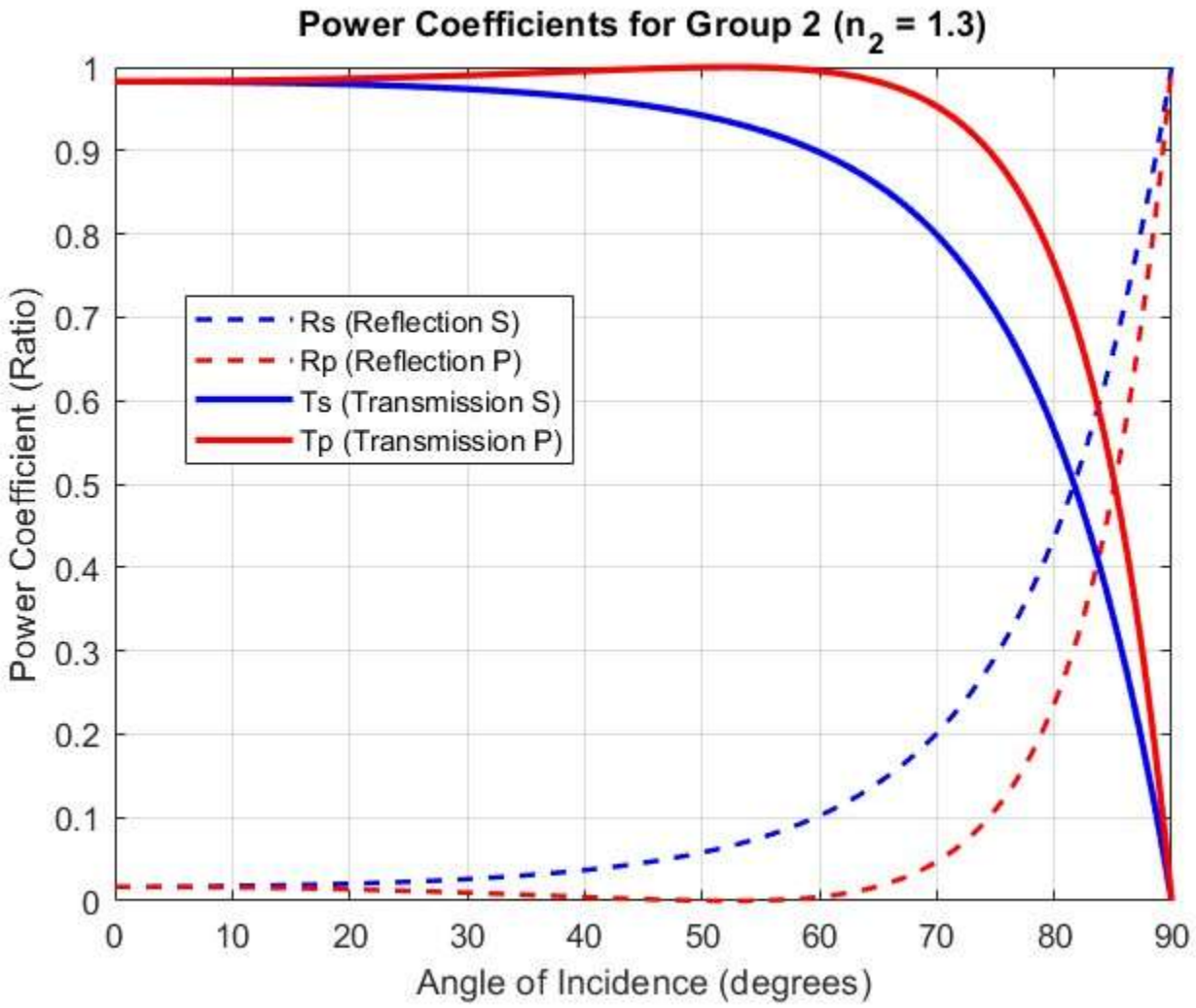
%    Compute Transmission Coefficients (T = 1 - R)
Ts = 1 - Rs;
Tp = 1 - Rp;

%    Plot
figure;
plot(theta_deg, Rs, 'b--', 'LineWidth', 1.5); hold on; % S-Reflect (Blue Dashed)
plot(theta_deg, Rp, 'r--', 'LineWidth', 1.5);         % P-Reflect (Red Dashed)
plot(theta_deg, Ts, 'b-', 'LineWidth', 2);            % S-Transmit (Blue Solid)
plot(theta_deg, Tp, 'r-', 'LineWidth', 2);            % P-Transmit (Red Solid)

% Formatting
title(['Power Coefficients for Group 2 (n_2 = ' num2str(n2) ')']);
xlabel('Angle of Incidence (degrees)');
ylabel('Power Coefficient (Ratio)');
legend('Rs (Reflection S)', 'Rp (Reflection P)', ...
       'Ts (Transmission S)', 'Tp (Transmission P)', 'Location', 'best');
grid on;
axis([0 90 0 1]);

%    Output Brewster Angle
brewster_angle = rad2deg(atan(n2/n1));
fprintf('Brewster Angle for n=%.2f is %.2f degrees.\n', n2, brewster_angle);
```

Plotted output curve:



Does the Brewster angle change as expected?

- **Standard (n=1.5):** $\theta_B \approx 56.3^\circ$
- **Group 2 (n=1.3):** $\theta_B \approx 52.4^\circ$
- **Conclusion:** Yes, as the refractive index decreases from 1.5 to 1.3, the Brewster angle shifts to a lower angle (θ_B decreases). This is consistent with the relation $\tan \theta_B = n_2/n_1$.